Indian Institute of Technology Gandhinagar

ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026

Tutorial-3 (Thevenin/Norton, Maximum Power Transfer, Inductor and Capacitor Response)

The goal of this tutorial is to master circuit simplification techniques for optimizing power delivery and analysing the fundamental behaviour of energy storage elements (e.g. inductor, capacitor) rather than just the dissipating elements (e.g. resistors).

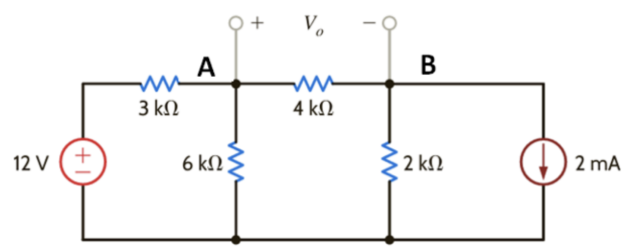
Q1. A sealed "Black Box" contains a linear resistive circuit with independent sources. You perform two measurements at the output terminals A-B:

1. Open Circuit Test: You measure a voltage of 24 V across terminals A-B.
2. Loaded Test: When you connect a 10 Ω resistor across terminals A-B, the voltage drops to 16 V.

a) Determine the Thevenin Equivalent Voltage (V_{th}) and Thevenin Resistance (R_{th}) of the black box.

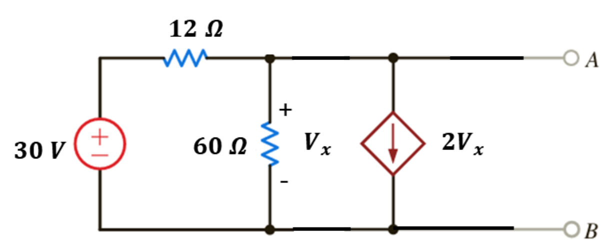
b) Draw the Norton Equivalent circuit for this box.

c) Calculate the Short Circuit current (I_{sc}) that would flow if you connected a wire directly between A and B.



Q2. You are given a representative circuit in a digital weighing scale. The left side of the node A creates a steady "Reference Voltage at node A" used for comparison. The right side of the node B connects to the weight sensor (here modeled as 2 mA current source with 2 kΩ resistance). To display the weight, the computer measures the difference in voltage between the reference and the sensor ($V_o = V_A - V_B$).

Find V_o (across the 4 kΩ load). You may choose Thevenin or Norton Equivalent circuit to calculate it.



Q3. You are trying to interface a sensor with a Feedback Control Module (the circuit shown). The module is a Black Box with terminals A and B. But, inside it contains an Active Feedback Element which monitors the output voltage V_x and draws a current equal to $2V_x$. To safely connect your sensor, you must determine the input resistance (R_{in} or R_{th}) of this module while it is active.

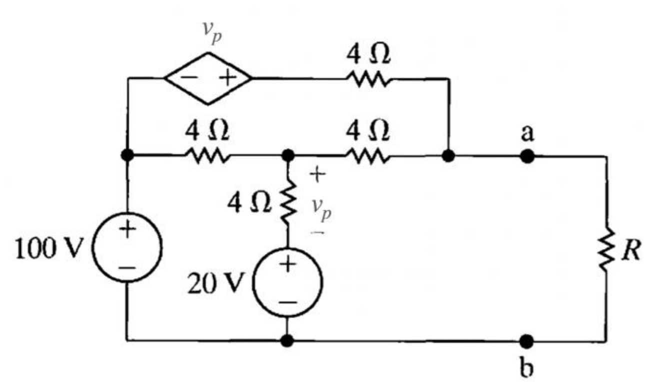
A. If one of you attempt to find R_{th} by turning off the independent 30 V source (shorting it) and turning off the dependent source (opening it), leaving only the resistors. What resistance value would you calculate
B. Keeping the dependent source active, reduce the given circuit into Thevenin equivalent circuit with respect of terminals *A* and *B*. What R_{th} would you calculate?
C. Next, you apply a **1 Volt Test Source** (V_{test}) at terminals A-B and measure the current (I_{test}) drawing from your test source. What R_{in} (V_{test}/I_{test}) would you calculate?
D. Compare R_{in} or R_{th} obtained for all three cases. Which are correct?

Q4. Three of you are testing identical 12 V chemical batteries for a new drone. Each of you connects a different load i.e. 2 Ω, 6Ω, and 18 Ω, respectively to their battery and measure the respective power outputs i.e. 4.5 W, 6 W, and 4.5 W to see which setup delivers the most energy.

A argues: *"I used the lowest resistance to let the most current flow. I should have the highest power (I^2R), but I don't!"* C argues: *"I used the highest resistance to get the biggest voltage drop. I should have the highest power (V^2/R), but I don't!"* *B argues: "I think that I just got lucky, but I don't why"*

i. Why are both A and C wrong? Explain why simply lowering or raising resistance doesn't always increase power.
ii. Based on B's "lucky" maximum value, can you calculate the internal resistance (R_{Th}) of the batteries?
iii. To understand *why* B won, A) Draw I_L vs R_L and, then P_L vs R_L. B) Alternatively,
   - Draw the I_L-V_L line for the battery with varying (R_L i.e. V_L on the x axis).
   - Mark the (V_L , I_L) points for A, B, and C on the plot.
   - Draw the rectangle/square for these points w.r.t origin. Which rectangle has the maximum area?

Q5. You are analysing the schematic of a high-voltage power supply unit (PSU) designed for a radar system. The dependent voltage source in the top branch represents an active feedback element, which reads the voltage across the internal 4Ω sense resistor and injects a boosting voltage equal to v_p into the output.



i. Find the value of R that enables the circuit shown to deliver maximum power to the terminals a, b.
ii. Find the maximum power delivered to R.
iii. If the radar load requires 1000 W of power at a voltage of 100 V, what value of load resistance R should be connected?

Q6. Unlike resistors which act instantly, real systems have "inertia" or "memory”. We use two special electrical components to model this "time delay":

1. The Capacitor (C): Stores energy in an Electric Field. It acts like a Spring or Tank. It fights changes in Voltage. Initially, it's an empty hole (easy to fill). Eventually, it becomes a solid wall (full).
2. The Inductor (L): Stores energy in a Magnetic Field. It acts like a Heavy Flywheel. It fights changes in Current. Initially, it's hard to push (heavy). Eventually, it spins freely (no friction).

You are analyzing the safety circuit for a heavy electronic vault door. The system uses two critical timing components to manage the lock solenoid (which unlocks the door) and the motor (which swings the door open). The two main circuits inside an automatic door system (Both use a 10V Source and a 1 kΩ Resistor):

1. Circuit A: The door must wait 2 seconds after the button is pressed before it unlocks. This circuit uses a Capacitor (C=1000 μF) in series to create that voltage delay.
2. Circuit B: The heavy motor needs to start spinning slowly to avoid jerking. This circuit uses an Inductor (L=1 H) in series to force the current to rise slowly ("Soft Start").

i. The Moment of Switch On (t=0): Calculate the current I flowing in each circuit the instant the switch is closed. Which circuit has zero current at the start?
ii. The Steady State (t=∞): Calculate the current I after the switch has been closed for a long time.
iii. Calculate its time constant τ of circuit A. Does this value meet the requirement?

More Practice Problems

- Irwin J.D. and Nelms R.M., Basic Engineering Circuit Analysis.
- Nilsson J., S.-Riedel, Electric-Circuits, 9th-ed, Prentice-Hall-2011.